
SVDppAG: Enhancing SVD++ with Attention and Gating

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Abstract

SVD++ is a popular collaborative filtering technique using both explicit ratings and implicit user feedback. Its simple aggregation and fusion of implicit feedback, however, treats all implicit interactions equally. We propose SVDppAG, an extension of SVD++ that adds an attention mechanism to adaptively weight the implicit user items by relevance and a gating mechanism for adaptive fusion of the explicit user signal with the implicit attended item signal. Our experiments show that SVDppAG achieves a lower Root Mean Square Error than SVD++ and other baselines. An ablation study shows that while both the attention and gating mechanisms, are effective enhancements, their combination does not yield further improvement.

1. Introduction

Recommender systems help users discover items matching their preferences from vast user data (Ricci et al., 2010). Collaborative Filtering (CF) is a popular approach which makes use of past user-item interactions to predict the future preferences (Su & Khoshgoftaar, 2009). Interactions can be in explicit form, such as rating an item, or implicit form, like saving an item for later, which shows interest without guarantee to consume and rate it ever. Matrix Factorization (MF) techniques, learning latent representations for users and items, have proven to be effective in CF (Koren et al., 2009). SVD++, a popular MF method, combines explicit user factors with aggregated latent factors from implicitly interacted items for accurate predictions. (Koren, 2008).

While SVD++ offers a strong baseline, its original formulation has seen extensions, for instance, by incorporating temporal dynamics (Koren, 2010). Moreover, its aggregation step treats all implicit interactions as equally important after normalization. The following fusion of the aggregated implicit signal with the explicit factor of the given user is

a simple sum. We argue that the aggregation can not capture the varying relevance of different implicit items and the fusion step lacks a mechanism to individually adjust the overall importance of the implicit term in the final user representation. Inspired by the success of attention mechanisms in other areas of CF, such as dynamically weighting historical items (He et al., 2018), and the potential of gating for adaptive signal combination, we aim to enhance these aspects of SVD++.

To address these limitations, we propose SVD++ with Attention and Gating (SVDppAG). SVDppAG introduces two improvements:

1. An attention mechanism to dynamically adjust the importance of an implicit interaction by a given user.
2. A gating mechanism which enables the model to learn an adaptive weighting between the explicit and implicit signal in the fusion step.

We expect that this will enable the model to learn relationships between user’s and their relevant implicit interactions and also how much weight to give them overall.

The remainder of this paper is structured as follows: Section 2 details the SVDppAG model. Section 3 introduces the experimental setup and Section 4 presents the results. Finally, Section 5 discusses the findings and Section 6 concludes the paper.

2. Proposed Method

The proposed SVD++ with Attention and Gating (SVDppAG) model extends the standard SVD++ model by adding an attention mechanism for adaptive implicit feedback aggregation and a gating mechanism for the adaptive fusion of explicit and implicit signals.

2.1. SVD++

The SVD++ rating prediction $\hat{r}_{ui}$ for user u and item i is given by:

$$\hat{r}_{ui} = \mu + b_u + b_i + q_i^T \left(p_u + |N(u)|^{-\frac{1}{2}} \sum_{j \in N(u)} y_j \right) \quad (1)$$

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where μ is the global rating mean, b_u and b_i are user and item biases, $q_i, p_u, y_j \in \mathbb{R}^d$ are d -dimensional latent factors for item i , user u and implicitly interacted item j , respectively. $N(u)$ is the set of implicit interactions of user u .

2.2. SVDppAG

SVDppAG improves two aspects of the SVD++ model: The aggregation of the implicit signal and the subsequent fusion with the user’s explicit factor. The two novelties are introduced in the following.

2.2.1. ATTENTION-BASED AGGREGATION

Instead of a simple normalized sum aggregation, SVDppAG uses an attention mechanism to dynamically weigh each implicit item y_j based on its relevance to a given user’s preference factor p_u . This enables to focus on relevant implicit interactions. The user preference factor p_u is used as query and the implicit factors $y_j \in N(u)$ as keys and values. The aggregation works as follows:

1. **Attention Score:** For each implicit interacted item $j \in N(u)$ of user u , its score is calculated by:

$$s_{uj} := \frac{p_u^T y_j}{\sqrt{d}} \quad (2)$$

The scaling factor $\frac{1}{\sqrt{d}}$ helps to prevent issues with large dot product magnitudes, as commonly used in attention mechanisms (Bahdanau et al., 2016; Vaswani et al., 2017).

2. **Attention Weights:** The attention scores s_{uj} are normalized using a softmax function to receive attention weights:

$$\alpha_{uj} := \frac{\exp(s_{uj})}{\sum_{k \in N(u)} \exp(s_{uk})} \quad (3)$$

3. **Attended Implicit Representation:** The final implicit representation $\hat{y}_u$ for user u is then given by the attention weighted sum of implicit representations:

$$\hat{y}_u := \sum_{j \in N(u)} \alpha_{uj} y_j \quad (4)$$

Note that $\hat{y}_u$ is the zero vector if $|N(u)| = 0$, i.e. user u has no implicit interactions.

2.2.2. GATED FUSION OF USER SIGNAL

For the fusion step a gating mechanism is introduced that learns an adaptive weighting of the user factor p_u with the attended implicit signal $\hat{y}_u$. The gating value $g_u \in (0, 1)$ is calculated by concatenating $p_u, \hat{y}_u$ into a single vector followed by an affine transformation and a sigmoid activation $\sigma(\cdot)$:

$$g_u = \sigma(W[p_u; \hat{y}_u] + b) \quad (5)$$

where $W \in \mathbb{R}^{1 \times 2d}$ and $b \in \mathbb{R}$ are learnable parameters. The gating value g_u then steers the influence of each component to get the final user representation $\hat{p}_u$:

$$\hat{p}_u := g_u \cdot p_u + (1 - g_u) \cdot \hat{y}_u \quad (6)$$

This formulation allows the model to learn the optimal influence of the implicit signal. This gate introduces $2d + 1$ additional learnable parameters.

2.3. Prediction and Training

The SVDppAG prediction for a rating is then given by:

$$\hat{r}_{ui} = \mu + b_u + b_i + q_i^T \hat{p}_u \quad (7)$$

The model is trained by minimizing the Mean Squared Error (MSE) between predicted and actual rating together with L2 regularization on all learnable parameters θ . The loss function L is given by:

$$L = \frac{1}{|D_{\text{train}}|} \sum_{(u,i,r_{ui}) \in D_{\text{train}}} (\hat{r}_{ui} - r_{ui})^2 + \lambda \|\theta\|_F^2 \quad (8)$$

where D is the set of explicit training data, $\hat{r}_{ui}$ is the predicted rating, r_{ui} is the true rating and λ is the regularization strength hyperparameter.

3. Experimental Setup

This section details the experiments used to evaluate the SVDppAG model and compare it against two baselines. We introduce the dataset, baseline models, evaluation metrics and the hyperparameter tuning process.

3.1. Dataset and Preprocessing

Our experiments were conducted on the Collaborative Filtering dataset provided by the Computation Intelligence Lab course, consisting of:

- 10,000 scientists (users) and 1,000 papers (items).
- 1,128,187 explicit interactions (ratings) in the range of 1 to 5 stars.
- 328,839 implicit interactions.

The interaction matrix thus has a density of $\approx 11\%$ explicit interactions and $\approx 3\%$ implicit interactions. The mean rating is ≈ 3.82 with a standard deviation of ≈ 1.01 . The number of implicit interactions per user varies significantly, with a mean of 32.88, ranging from 0 to 95.

We split the data into training, validation, and test sets using a random 80%/10%/10% split based on interactions. The global rating mean μ , used in the SVD++ and SVDppAG models, was pre-calculated as the average rating from the training data. No other data preprocessing steps were applied.

3.2. Baseline Models

We compare SVDppAG against the following baseline models:

- **SVD++:** Our implementation of the original SVD++ model as presented in (Koren, 2008).
- **Alternating least square (ALS):** An implementation of the ALS algorithm (Hu et al., 2008) which only leverages explicit interactions. Implicit interactions were discarded for its training.

3.3. Evaluation Metrics

The model performance is evaluated using the Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{|D_{\text{test}}|} \sum_{(u,i,r_{ui}) \in D_{\text{test}}} (\hat{r}_{ui} - r_{ui})^2} \quad (9)$$

where D_{test} is the test set, r_{ui} is the true rating, and $\hat{r}_{ui}$ is the predicted rating. Lower RMSE values indicate better accuracy.

3.4. Hyperparameter Tuning and Training

Hyperparameter tuning was applied for all models. Tuned hyperparameters were embedding dimension (d), L2 regularization strength (λ), and learning rate (LR) for the gradient-based methods. For SVD++ and SVDppAG, explored embedding dimensions ranged from 4 to 256, L2 regularization from 0.005 to 0.04, and learning rates from 10^{-4} to 10^{-2} . For ALS, tuning only included embedding dimension and L2 regularization. Gradient-based models, SVD++ and SVDppAG, used the Adam optimizer (Kingma & Ba, 2017) and trained for up to 40 epochs with early stopping (patience 5). A learning rate scheduler halved LR after 3 epochs without improvement. No run reached 40 epochs. The alternating optimization for ALS was performed for 50 iterations. The selected optimal hyperparameters for each model, along with their respective search spaces, are provided in the Appendix A. Gradient-based models are implemented in PyTorch (Paszke et al., 2019). The source code is publicly available.¹

4. Results and Analysis

This section presents the evaluation of our proposed SVDppAG model. We first compare its overall performance against the baseline models. Following, we conduct an ablation study to quantify the impact of the introduced components. Finally, we analyze the impact of the embedding dimension on the prediction accuracy.

¹<https://github.com/dubachti/SVDpp-Attention-Gating>

4.1. Overall Performance Comparison

To evaluate the performance of our proposed SVDppAG model, we compare its accuracy against SVD++ and ALS baselines. Each model was configured with its optimal hyperparameters, as determined on the validation set. Table 1 shows the achieved RMSE metric on the test set. To ensure robustness of our findings, the reported RMSE values consist of mean and standard deviation over 3 training runs with different random initializations.

Model	Test RMSE
ALS	$0.856 \pm 1.22\text{e-}4$
SVD++	$0.853 \pm 3.03\text{e-}4$
SVDppAG (Ours)	$0.844 \pm 3.23\text{e-}4$

Table 1. Overall performance comparison on the test set. RMSE values are mean $\pm$ standard deviation over 3 runs. The best performing model is highlighted in **bold**. Lower RMSE indicates better predictive accuracy.

The results demonstrate the improved performance of our SVDppAG model which achieved the lowest test RMSE of 0.844, outperforming both baseline models. Compared to the popular SVD++ method, SVDppAG provided an improvement of 0.009. The ALS model, which does not incorporate implicit feedback, provides the highest RMSE which demonstrates the importance of incorporating this type of feedback.

4.2. Ablation Study

To isolate and quantify the individual impact of the attention and gating mechanism of the SVDppAG model, we conduct an ablation study. The study compares the baseline SVD++ and the full SVDppAG model against two intermediate variants:

- **SVDppA (Attention Only):** SVD++ augmented solely with our proposed attention mechanism for implicit signal aggregation.
- **SVDppG (Gating Only):** SVD++ augmented solely with our proposed gating mechanism for fusing explicit and implicit user representations.

All models were configured with the optimal hyperparameters determined for the full SVDppAG model, hence the SVD++ accuracy differs here from the one in Table 1. The resulting test set RMSE, averaged over 3 runs, is presented in Table 2.

Introducing only the attention mechanism (SVDppA) reduced the RMSE by 1.3% from 0.855 to 0.844. Similarly, introducing only the gating mechanism (SVDppG) resulted in an RMSE of 0.844 as well. These findings confirm that both the attention-based implicit representation aggregation and the gated fusion of implicit and user representation

Model Configuration	Test RMSE
SVD++ (Attn: No, Gate: No)	$0.855 \pm 4.64\text{e-}4$
SVDppA (Attn: Yes, Gate: No)	$0.844 \pm 2.47\text{e-}4$
SVDppG (Attn: No, Gate: Yes)	$0.844 \pm 1.78\text{e-}4$
SVDppAG (Attn: Yes, Gate: Yes)	$0.844 \pm 3.23\text{e-}4$

Table 2. Ablation study results on the test set, showing the impact of attention (Attn) and gating (Gate) mechanisms. RMSE values are mean $\pm$ standard deviation over 3 runs. Lower RMSE indicates better performance.

contribute positively to the model’s accuracy. Interestingly, the full SVDppAG model achieves again the same RMSE by combining the attention and the gating mechanism as the SVDppA or SVDppG models do. While there were variations in further decimal places, these differences were well within the standard deviation of the runs. This indicates that there is no additional benefit from combining both mechanisms in this setup.

4.3. Impact of Latent Representation Dimension

To understand how the model complexity, in particular the dimensionality of the latent factors affect the performance, we analyzed the RMSE of the SVD++ baseline and our SVDppAG model across a range of embedding dimensions. Note that for each tested dimension, the other hyperparameters were all tuned to their optimal value using the validation set. The results are presented in Figure 1.

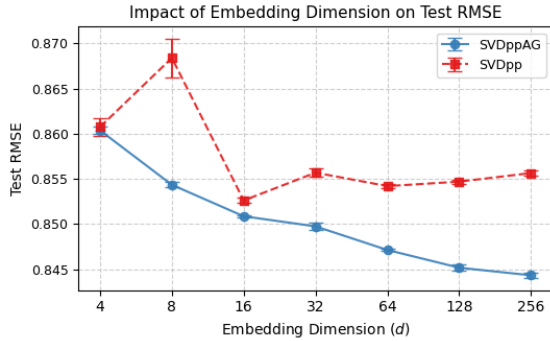


Figure 1. Test set RMSE as a function of the latent representation dimension (d) for SVDppAG and SVD++. For each dimension, models were configured with their respective optimal learning rates and L2 regularization strengths.

As seen in Figure 1, SVD++ reaches its lowest RMSE already at embedding dimensions $d = 16$ (RMSE: 0.853), after which it slowly increases again, a common sign of overfitting with growing model capacity. In contrast, SVDppAG, with an embedding dimension of $d \geq 8$, consistently outperforms SVD++ and shows consistent reduction in RMSE with growing d , reaching the best value within our tested parameters at the maximal embedding dimension of $d = 256$

(RMSE: 0.844). This hints at SVDppAG being able to leverage more complex representations.

5. Discussion

Our work aimed to improve the SVD++ model by incorporating attention and gating mechanisms, resulting in our SVDppAG model. The results presented in Section 4 offer insights into the contribution of the changes.

SVDppAG consistently outperformed the SVD++ and ALS baselines (Table 1). This supports our hypothesis that adaptive processing of implicit feedback improves accuracy. Notably, SVDppAG also demonstrated better utilization of higher embedding dimensions compared to SVD++, which showed signs of overfitting (Figure 1).

The ablation study (Table 2) provided deeper insight into the individual model components. Both the attention-only (SVDppA) and gating-only (SVDppG) variants individually showed significant RMSE reductions (approx. 1.3%) compared to an identically configured SVD++ baseline. This shows that both components are effective when used standalone for improving SVD++ prediction accuracy.

A key finding, however, was that the full SVDppAG model offered no further RMSE reduction over SVDppA or SVDppG variants. The performance of all 3 variants was nearly identical. This suggests that there is no compositional benefit when using both components together. This may indicate that, for this dataset, either the attention or gating mechanism is already sufficient to leverage the potential of the refined implicit signal, with the second mechanism offering no additional explanatory power. This implies that while both attention and gating are good additions to SVD++, using just one of those components might be enough to achieve the performance gains while using a simpler model.

Limitations include evaluation on a single dataset, missing study data sparsity impact study, and exploration only specific attention and gating architectures. The computational overhead of these components was also not benchmarked.

6. Conclusion and Future Work

In conclusion, this work demonstrates that both attention and gating mechanisms are powerful individual improvements for the SVD++ model. While their direct combination did not show further benefits in our dataset, the strong independent performance of the individual components is promising for developing more effective collaborative filtering models. Future work could focus on understanding the relationship between the attention and gating mechanism more deeply, potentially by analyzing the learned parameters. Evaluation on different datasets would also be interesting to further test the generalization of the approach.

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A. Optimal Hyperparameters

This section shows the optimal hyperparameters selected for each model based on the validation set performance, along with the search ranges explored during grid search. Table 3 presents these details.

Model	Emb. Dim. (d)	L2 Reg. (λ)	Learning Rate (LR)
ALS	12	20	N/A
Search Range	{4, 8, 12, 16, 20, 24}	{16, 18, 20, 22, 24}	—
SVD++	16	0.02	0.01
Search Range	{4, 16, 32, 64, 128, 256}	{0.005, 0.01, 0.02, 0.04}	$\{10^{-4}, 10^{-3}, 10^{-2}\}$
SVDppAG (and ablations)	256	0.01	0.001
Search Range	{4, 16, 32, 64, 128, 256}	{0.005, 0.01, 0.02, 0.04}	$\{10^{-4}, 10^{-3}, 10^{-2}\}$

Table 3. Selected optimal hyperparameters and explored search ranges for all evaluated models. Optimal values are shown in **bold**; search ranges are shown in {brackets}.